

Flash Memory Wear-Leveling Strategy

ELIoT-RTU

Background

The device stores persistent data in a NOR flash memory chip — a type of non-volatile storage that retains values without power. Unlike RAM, flash memory has a physical limitation: each storage location can only be erased and rewritten a finite number of times. The manufacturer of this chip guarantees **100,000 erase-write cycles** per storage location before reliability begins to degrade.

The firmware saves configuration and runtime values to this chip on a regular interval. Without protection, frequent writes would exhaust the chip's rated cycles within months.

How the Wear Leveling System Works

Core Concept — Sequential Sub-Slots

Rather than erasing and rewriting the same physical location on every save, each logical storage address is backed by a sector of flash memory divided into **256 sub-slots** of 16 bytes each.

Writes advance sequentially through these sub-slots:

```
Save #1    -> written to sub-slot 0
Save #2    -> written to sub-slot 1
Save #3    -> written to sub-slot 2
...
Save #256  -> written to sub-slot 255
|
v
Sector erased -> back to sub-slot 0
```

The sector is erased only once every 256 saves, reducing erase cycle consumption by a factor of 256 compared to erasing on every write.

Power-Failure Safety

Each sub-slot is committed with a two-step sequence:

1. The value is written first
2. A “valid” marker is written last

If the device loses power between these two steps, the marker is absent and the partial write is ignored on the next boot. The previous valid value remains intact and readable.

Duplicate Suppression

Before committing a new value, the system reads back the currently stored value and compares. If the value has not changed, the write is skipped entirely — no sub-slot is consumed and no wear occurs. This is effective whenever a periodic save timer fires but the underlying value has not changed since the last save.

Lifespan Calculation

Assumptions

Parameter	Value
Chip endurance rating	100,000 erase cycles per location
Practical target (80% safety margin)	80,000 erase cycles
Sub-slots per location	256
Write interval	Every 30 seconds
Daily operating hours	12 hours

The 12-hour daily operating figure reflects the ELIoT-RTU's power source: the unit is solar-powered, and generation — and therefore active operation — is only available during the roughly 12 hours of daylight each day. The device is not active and does not perform periodic writes outside this window.

Writes Per Day

```
Writes per hour = 3600 seconds / 30 seconds = 120
Writes per day  = 120 x 12 hours = 1,440 writes/day
```

Erase Cycles Per Day

```
Erase cycles/day = 1,440 writes / 256 sub-slots = 5.625 cycles/day
```

Time to Reach 80% of Rated Endurance

Days = $80,000 / 5.625 = 14,222$ days
Years = $14,222 / 365 \approx 38.9$ years

Under continuous 12-hour daily writes with no duplicate suppression, the current wear-leveling strategy reaches 80% of the chip's rated endurance in approximately **39 years** — well beyond the expected operational lifetime of the device hardware.

Effect of Duplicate Suppression

In real-world operation, the stored value will not change on every timer fire. The table below shows how lifespan scales with the rate of actual change.

Value changes every...	Effective writes/day	Years to 80% limit
30 seconds (every write)	1,440	39 years
1 minute	720	78 years
5 minutes	144	389 years
10 minutes	72	778 years

Summary

The current wear-leveling strategy — 256 sequential sub-slots per storage location with a power-safe two-step commit — reduces erase-cycle consumption to roughly 5.6 cycles per day under continuous 12-hour daily writes, giving an estimated 39 years before reaching 80% of the chip's rated endurance. Combined with duplicate suppression — which skips writes entirely when the value has not changed — the chip is protected well beyond the operational lifetime of the device hardware.

No maintenance or configuration is required. The system operates transparently within the firmware.